Muhammad Rayyan Ahmed

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M.Sc. Data Science (in progress), TU Ilmenau | B.S. Artificial Intelligence | Four SAP Certified Associate certifications

Cover Letter: Student Assistant (HiWi) for AI-Assisted 3D Segmentation of Photonic Nanostructures

Max Planck Institute for Intelligent Systems Stuttgart 2026-10-23

Dear Max Planck Institute for Intelligent Systems team,

I am applying for the Student Assistant position in AI-assisted 3D segmentation of photonic nanostructures. Applying machine learning to scientific images, and doing it carefully enough that the results can be trusted for research, is the kind of computer vision work I enjoy most.

I am an M.Sc. Data Science student at TU Ilmenau with a B.S. in Artificial Intelligence, where I built computer vision and NLP projects in PyTorch. I also bring industry experience in SQL, data pipelines and dashboards from SAP data & analytics work.

- **Image-based deep learning,** I built a PyTorch retinal OCT classifier on EfficientNet-B0 reaching 94.11% accuracy on a 968-image test set, including all preprocessing and augmentation
- **Model evaluation,** in my thesis team I created training data, evaluated and tuned an EfficientNet-B0 model and compared EasyOCR configurations
- **Careful documentation,** I documented data sources, calculations and results for every dashboard I built, so others could reproduce and support them

I would bring careful, consistent work on image data, experience applying and tuning deep learning models, and the habit of documenting workflows so analyses are reproducible. I have not used Amira yet, and I appreciate that training is provided.

I am enrolled at TU Ilmenau, available immediately for up to 20 hours per week, and happy to relocate to Stuttgart. I would be glad to discuss how I can contribute.

I work in English (IELTS 8.0) and am currently learning German (A1).